

Chemical characterization of PM_{2.5} collected from a rural coastal island of the Bay of Bengal (Bhola, Bangladesh)

Mohammad Shohel^{1,2} · Magdalena Kistler³ · Mohammad Arifur Rahman⁴ · Anne Kasper-Giebl³ · Jeffrey S. Reid⁵ · Abdus Salam¹

Received: 17 July 2017 / Accepted: 6 November 2017 / Published online: 30 November 2017
© Springer-Verlag GmbH Germany, part of Springer Nature 2017

Abstract This work focuses on the chemical characterization of fine aerosol particles (PM_{2.5}) collected from a rural remote island of the Bay of Bengal (Bhola, Bangladesh) from April to August, 2013. PM_{2.5} particle-loaded filters were analyzed for organic carbon (OC), elemental carbon (EC), water-soluble ions, and selected saccharides (levoglucosan, mannosan, galactosan, arabitol, and mannitol). The average PM_{2.5} mass was $15.0 \pm 6.9 \mu\text{g m}^{-3}$. Organic carbon and elemental carbon comprised roughly half of the analyzed components. Organic carbon was the predominant contributor to total carbon (TC) and accounting for about 28% of PM_{2.5} mass. Secondary organic carbon (SOC) was inferred to be ~26% of OC. The sum of ions comprised ~27% of PM_{2.5} mass. The contribution of sea salt aerosol was smaller than expected for a sea-near site (17%), and very high chloride depletion was observed (78%). NssSO_4^{2-} was a dominant ionic component with an average

concentration of $2.0 \mu\text{g m}^{-3}$ followed by Na^+ , NH_4^+ , and nssCa^{2+} . The average concentration of arabitol and mannitol was 0.11 and $0.14 \mu\text{g m}^{-3}$, respectively, while levoglucosan and its stereoisomers (mannosan and galactosan) were below detection limit. $\text{NH}_4^+/\text{SO}_4^{2-}$ equivalent ratio was 0.30 ± 0.13 indicating that secondary inorganic aerosol is not the main source of SO_4^{2-} . Enrichment factor (EF) analysis showed that SO_4^{2-} and NO_3^- were enriched in atmospheric particles compared to sea aerosol and soil indicating their anthropogenic origin. Higher OC/EC ratio (3.70 ± 0.88) was a good indicator of the secondary organic compounds formation. Other ratios (OC/EC, K^+/EC , $\text{nssSO}_4^{2-}/\text{EC}$) and correlation analysis suggested mixed sources for carbonaceous components. Arabitol and mannitol both showed strong correlation with EC having R^2 value 0.89 and 0.95, respectively. Air mass trajectories analysis showed that concentrations of soil and anthropogenic species were lower for air masses originating from the sea (May–August) and were higher when air came from land (April).

Responsible editor: Gerhard Lammel

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s11356-017-0695-6>) contains supplementary material, which is available to authorized users.

✉ Abdus Salam
asalam@gmail.com

- ¹ Department of Chemistry, Faculty of Science, University of Dhaka, Dhaka 1000, Bangladesh
- ² Department of Chemistry, University of Iowa, Iowa City, IA 52242, USA
- ³ Institute of Chemical Technologies and Analytics, Vienna University of Technology, 1060 Vienna, Austria
- ⁴ Center for Air Resources Engineering and Science, Clarkson University, Potsdam, NY, USA
- ⁵ Naval Research Laboratory, 7 Grace Hopper Ave., Stop 2, Monterey, CA 93943, USA

Keywords PM_{2.5} · Sulfate · Elemental carbon · Saccharides · Organic carbon

Introduction

Aerosol particles in the atmosphere have an important role to govern climate, visibility, and human health. Outdoor air pollution has been linked to 3.7 million deaths in 2012, and the WHO classifies it currently as the world's largest single environmental health risk (WHO 2014). Emissions of aerosol particles and gases and the production of secondary pollutants resulting from chemical reactions in the atmosphere degrade the air quality significantly. Urban air quality has been profoundly deteriorated due to growth of urbanization, industrialization, and economy (Wahid 2006; Mondol et al. 2014).

The Indo-Gangetic Plain (IGP) encompasses 21% of the Indian subcontinent and hosts over 1/8th of the world's population (Shohel et al. 2017). It consists of the northern and eastern part of India, eastern parts of Pakistan, and Bangladesh (Taneja et al. 2014). The rapidly expanding economy of the IGP results a wide range of anthropogenic emissions. This region is an emitter of diverse anthropogenic aerosol such as soot, nitrates, sulfates, trace metals, and organic aerosols. Rapidly growing mega cities in this region are confronted with the great challenge of controlling air pollution (Chattopadhyay et al. 2010). Surface particulate matter network (SPARTAN) reported a higher concentration of $PM_{2.5}$ in the IGP region (e.g., Dhaka, $51.9 \mu\text{g m}^{-3}$; Kanpur, $99.3 \mu\text{g m}^{-3}$) compared to other sites around the world (e.g., Hanoi, $39.4 \mu\text{g m}^{-3}$; Bandung, $31.4 \mu\text{g m}^{-3}$; Atlanta, $9.1 \mu\text{g m}^{-3}$; Buenos Aires, $10.1 \mu\text{g m}^{-3}$; Rehovot, $16.1 \mu\text{g m}^{-3}$; Ilorin, $15.7 \mu\text{g m}^{-3}$; Pretoria, $6.4 \mu\text{g m}^{-3}$) (Snider et al. 2016). Higher concentrations of NO_3^- , SO_4^{2-} , black carbon, and various trace element oxides in the IGP region were also observed (Snider et al. 2016). The concentrations of ambient particles are strongly affected by long-range transport, very similar for the large region part and by regional enrichment which is to a large extent resulting from biomass burning and agricultural fires (Sinha et al. 2014). The meteorological seasons typical in the region determine the pollutants transport lowering the particle concentrations during the monsoon season (Pawar et al. 2015).

The air pollution situation in Bangladesh is extremely poor due to abundance and mismanagement of emission sources, e.g., traffic, brick kilns, industries, construction activities, and biomass burning (Salam et al. 2008; Snider et al. 2016). Moreover, due to its location in the Indo-Gangetic Plain, Bangladesh is also affected by transboundary air pollution (Begum et al. 2010, 2012). Although the air pollution in South Asia has been receiving attention during the past few decades, only a handful of studies have been conducted regarding aerosol characterization in Bangladesh. Among those, most were done in urban areas (e.g., Salam et al. 2003a, 2008, 2012; Begum et al. 2004; Rana et al. 2016). These studies report mainly PM levels in the megacities, e.g., Dhaka where the previously mentioned phenomena can be still observed with average $PM_{2.5}$ concentrations in the long-term sampling period from 2012 to 2015 between 240.3 and $327.4 \mu\text{g m}^{-3}$ (Rana et al. 2016). Less attention has been given to the rural areas and background, despite the fact that rural areas might pose an important source of atmospheric pollution (Saud et al. 2011; Tiwari et al. 2016). Our recent study in this important air quality monitoring location has revealed higher enrichment of anthropogenic species in dew water (Shohel et al. 2017). The first attempt for characterizing of aerosol particles at this remote rural island Bhola, Bangladesh, was made with a low-volume sampler (TSP) during May 14–21, 2001 (Salam et al. 2003b). According to Salam et al. (2003b) Bhola Island is strongly affected by emissions from urbanized regions of Southeast Asia.

Our current study concerns of the chemical composition of $PM_{2.5}$ at the coastal island Bhola, Bangladesh, during days selected out of the period between April and August, 2013, 12 years after the initial study. As Bhola Island can be considered a regional background site for IGP outflow, we emphasize the water-soluble ions, carbonaceous species, and saccharides in order to describe the prevailing aerosol types (natural or anthropogenic). The observed concentrations and relations of those chemical species are discussed with the help of correlation, enrichment factor, ratios analysis, and air mass trajectories.

Experimental

Description and meteorology of the sampling location

Bhola is a delta island of the Bay of Bengal with area 1441 km and population 1.7 million (Fig. 1). With its prevailing wintertime NW wind direction, the site is an important receptor for monitoring the outflow of IGP dense pollutants into the Bay of Bengal. The main occupation of the people of this island is agriculture and fishing. Local influences from traffic and industry are very low. Agricultural activities and biomass burning for cooking are the main local emission sources.

The metrological year at the sampling site can be divided into four different seasons: pre-monsoon (March–May), monsoon (June–September), post monsoon (October–November), and winter (December–February) (as stated in detail by Salam et al. 2003b). In this study, the $PM_{2.5}$ particles were collected during pre-monsoon and monsoon times. The average annual temperature and precipitation in Bhola are 25.9°C and 230 mm, respectively. The temperature during the sampling period varied between 24 and 33°C with an average of 28°C . Monthly average relative humidity (RH%) for pre-monsoon sampling time was in between 71 and 77% and monsoon time was in between 83 and 85%.

Methods

$PM_{2.5}$ collection

$PM_{2.5}$ was collected between April and August, 2013. Nine daily-samples were considered for this study. The aerosol sampling was carried out using a low-volume fine particulate ($PM_{2.5}$) sampler (model Envirotech APM 550, Envirotech Instrument Pvt. Ltd., India). The sampler was installed on the top of a tower (9 m above the ground) in the south-east part of the island observatory (Fig. 1: $22^\circ 10' 01'' \text{N}$, $90^\circ 45' 00'' \text{E}$), which was set up with the help of the National Aeronautics and Space Administration (NASA) Aeronet project, Naval Research Laboratory (NRL), and Office of Naval Research Global (ONRG), USA. Quartz filters (Gelman, Membrane Filters, Type TISSU Quartz 2500QAT-UP, 47-

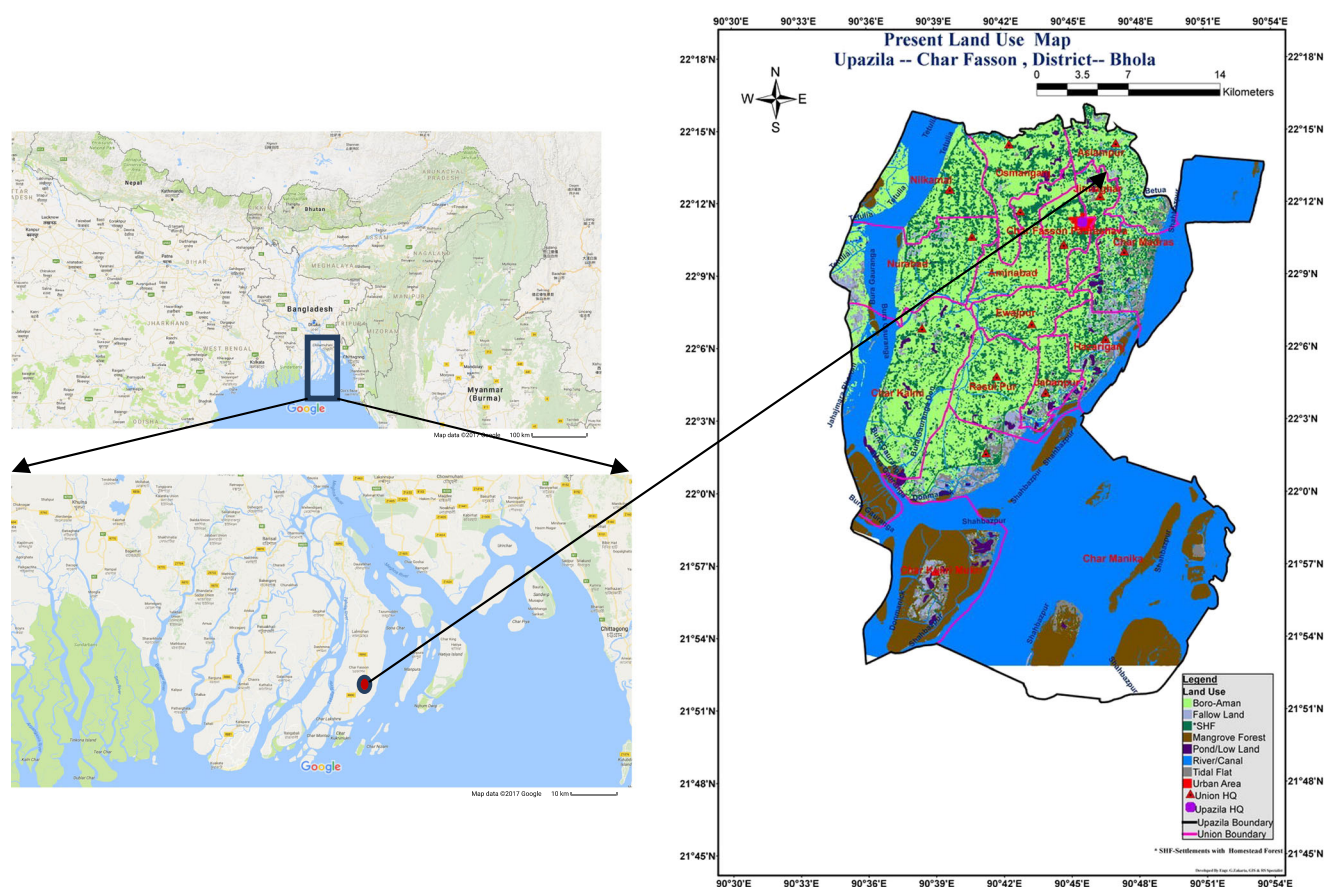


Fig. 1 Map of Bangladesh [left top panel], coastal island Bhola [left bottom panel]. Red circle approximates the sampling location (source Google Maps). Land use map of the sampling location [right panel] (source Ministry of Land, Governments of the Peoples Republic of Bangladesh)

mm diameter, Pall Corporation, USA) were equipped inside a sampler to collect particulate matter. The average air volume collected per sample was 14.0 m³. Blank corrections were done based on the analysis of field blank. The PM_{2.5}-loaded filters were stored in the refrigerator after sampling to minimize volatilization and evaporation. PM_{2.5}-loaded filters were equilibrated for 48 h at a controlled temperature (22 ± 2 °C) and relative humidity (55 ± 5% RH) prior to weighing before and after the sampling procedure. The filters were transported to Vienna University of Technology, Austria by air luggage for chemical analysis.

Inorganic ions (Cl⁻, NO₂⁻, NO₃⁻, SO₄²⁻, Na⁺, NH₄⁺, K⁺, Mg²⁺, Ca²⁺)

Inorganic ions were determined isocratic ion chromatography. A filter aliquot (1 × Ø10 mm, 0.8 cm²) was extracted in 3 mL of ultra-pure water (Milli-QPlus, 185, Millipore). After sonicating and centrifuging, the extract was analyzed by ion chromatography using a Thermo Scientific (previously Dionex) ICS 1100 system equipped with an AS22 anion-exchange column, an ASRS self-

regenerating suppressor, and a DS6 heated conductivity cell as a detector. The buffer containing 1.8 mm Na₂CO₃ and 1.7 mm NaHCO₃ was used as a mobile phase. Cations were extracted from filter aliquots (1 × Ø10 mm, 0.8 cm²) with the chromatography eluent, 38 mm methane sulphonic acid (MSA) to achieve a better solution of Ca²⁺ and Mg²⁺. After sonication and centrifugation, the extract is analyzed on a Dionex ICS 3000 system using a CS16 cation-exchange column, a self-regenerating suppressor (CERS), and a conductivity detector.

Na⁺ was used as a reference element for the source of sea water to differentiate sea salt (ss) from non-sea salt (nss) parts in aerosol and the following formula was used (Jusino-Atresino et al. 2016; Arimoto et al. 1996):

$$\text{nssC}_X = C_X(\text{aerosol}) - \left\{ (C_X/C_{\text{Na}})_{\text{sea}} \times (C_{\text{Na}})_{\text{aerosol}} \right\} \quad (1)$$

where: C_X (aerosol) is the concentration of the species X in aerosol sample, (C_X/C_{Na})_{sea} is the ratio of the given species X and Na⁺ in sea water, and (C_{Na})_{aerosol} is the concentration of Na⁺ in the aerosol sample. The sea salt aerosol concentration in aerosol was calculated using the formula (Bates et al. 2001; Jusino-Atresino et al. 2016).

Sea Salt Aerosol (SSA)

$$= \text{Cl}^-_{(\text{aerosol})} - (\text{Na}^+_{(\text{aerosol})} \times 1.47) \quad (2)$$

where, 1.47 is the mass ratio of $(\text{Na}^+ + \text{K}^+ + \text{Mg}^{2+} + \text{Ca}^{2+} + \text{SO}_4^{2-} + \text{HCO}_3^-)$ to Na^+ in seawater.

To calculate chloride depletion, the following equation was used assuming all Na^+ in aerosol was from sea water and the average ratio of Cl^- to Na^+ equals to 1.81 (Zhao and Gao 2008; Xu et al. 2013).

% Chloride Depletion

$$= \left[\frac{\{(1.81 \times \text{Na}^+_{\text{aerosol}}) - \text{Cl}^-_{\text{aerosol}}\}}{\{(1.81 \times \text{Na}^+_{\text{aerosol}})\}} \right] \times 100 \quad (3)$$

Carbonaceous species (OC and EC)

Thermal-optical determination of carbonaceous matter was conducted with a Lab OC-EC Aerosol Analyzer (Sunset Laboratory Inc.). The NIOSH 870 (National Institute for Occupational Safety and Health) temperature protocol and transmittance signal were used (Birch and Cary 1996).

Saccharides (levoglucosan, mannosan, galactosan, arabitol, mannitol)

The determination of saccharides was performed with an improved high-performance anion-exchange chromatography with pulsed amperometric detection (HPAEC-PAD) (Arfelli and Sartini 2014). This method is recommended for atmospherically relevant sugars (sugar alcohols, monosaccharides, and monosaccharide anhydrides) and widely described by Iinuma et al. (2009) and Arfelli and Sartini (2014). Before the analysis, a filter aliquot ($1 \times \text{Ø}10 \text{ mm}$, 0.8 cm^2) was eluted with 3 mL of ultra-pure water (Milli-QPlus, 185, Millipore), sonicated for 20 min, and subsequently centrifuged for 30 min. The extract was analyzed with an ion chromatography system (Dionex® ICS 3000), operated with a NaOH gradient (initial concentration, 480 mm NaOH; final concentration, 650 mm NaOH) run on a Carbo Pac MA1 column.

Quality assurance of the analytical methods

All analyses were conducted with respect to routine protocols worked out for the chemical characterization of particulate matter collected on quartz filters (as described in, e.g., Caseiro et al. 2009; Shahid et al. 2016; Styszko et al. 2017). In the chromatography methods, an appropriate calibration procedure was repeated at the beginning, in the middle, and at the end of a batch. The slope, intercept, and quality of fit of

the curve (in all cases $> 99\%$) were averaged over a batch, protocolled, and verified against continuous recording. The detection limits were calculated using the standard deviation of five-fold measurements of respective blank sample or lowest standard (in case when the field blank filter revealed no signals). Calibration of ions was performed with external mixed standards prepared from Merck® stock solutions. The limit of detection (expressed as three-fold standard deviation and calculated for the sampling conditions of this study) was at $0.02 \mu\text{g m}^{-3}$. The quantification of saccharides was carried out with external standards prepared from pure substances (Merck®, Fluka®). Detection limits for the analyzed anhydrous saccharides and sugar alcohols were similar and recalculated to the sample conditions equaled to $0.02 \mu\text{g m}^{-3}$. The quantification of carbonaceous fractions was based upon an internal procedure conducted newly for each measurement. A calibration gas (5% CH_4 in He) was automatically dosed into the FID detector. The accuracy of calibration factor was checked externally once a day against a Saccharose solution prepared from pure substance (Merck®). Detection limits of the method were $0.02 \mu\text{g m}^{-3}$ for elemental carbon and $0.28 \mu\text{g m}^{-3}$ for organic carbon. The accuracy of applied methods was assured by participation in the ring tests and inter-laboratory comparisons. The long-term reproducibility (between ring tests) was checked by regular analyses of reference samples.

Backward air mass trajectory analysis

Backward air mass trajectory analyses were carried out to identify the transport pattern of the air masses arriving at the sampling site of coastal Bhola (Salam et al. 2003b). The backward trajectories were calculated by the NOAA HYSPLIT model (Hybrid Single-Particle Lagrangian Integrated Trajectory Model) (<http://www.arl.noaa.gov/>). The frequency of the air mass trajectories were calculated of 30 days average at 300 m AGL (above ground level). Two main synoptic backward air mass trajectories were found—(i) air mass originating from the Bay of Bengal during monsoon and (ii) air mass originating from the IGP region during winter. Backward air mass trajectories of each individual sampling day were also calculated for 120 h to see the transport of air masses of each sampling day.

Results and discussion

Ionic composition and correlation analysis

The average concentration of $\text{PM}_{2.5}$ in the atmosphere of coastal Bhola during sampling period was $15.0 \pm 6.9 \mu\text{g m}^{-3}$ varied between 21.0 and $4.0 \mu\text{g m}^{-3}$. The difference of the $\text{PM}_{2.5}$ concentrations between the days in pre-monsoon and

monsoon seasons was $8.9 \mu\text{g m}^{-3}$. However, the concentrations of different chemical species for the size fraction of $\text{PM}_{2.5}$ are given in Table 1. Sodium and ammonium were major cations, and sulfate and nitrate were major anions in the $\text{PM}_{2.5}$ collected at Bhola Island (Table 1). The sum of ions comprised 27% of total $\text{PM}_{2.5}$ mass. The ionic balance based on the molar equivalent concentration ($\Sigma\text{cation}/\Sigma\text{anion}$) were on average = 1.0, and ranged between 0.65 and 1.70. The lowest value was observed for the sample characterized by lowest concentration of all components and thus might be burdened with higher uncertainty. The NIOSH870 Protocol applied for the determination of carbonaceous fractions allowed us to verify if a significant amount of carbonate carbon is present in the sample (visible as a very sharp peak during the highest temperature step in the measurement). This was not observed in any of the samples, so in case of the cation excess, we expect that organic anions such as oxalate may contribute to the organic fraction.

The percentage contribution of sea salt aerosol (SSA) with respect to total ionic mass during different sampling days was in between 6.94 and 47.21% with an average 17.17%. The Cl^- and Na^+ of 1.8 in sea water indicated a depletion of chloride (Jusino-Atresino et al. 2016). Chloride depletion in marine aerosols occurs due to reactions of sea salt with acidic compounds, such as nitric acid, sulfuric acid, and certain organic acids, resulting in the loss of Cl^- in the form of gaseous HCl (Virkkula et al. 2006; Pakkanen 1996; Xu et al. 2013). The percent of chloride depletion in the present study was in between 46.15 and 94.71% with an average 78.23%.

The sulfate to nitrate ratio is often used to indicate the possible sources of aerosol (Verma et al. 2010), although this approach may have some uncertainties (Arimoto et al. 1996), if the complex formation pathways of those secondary inorganic ions are considered. Higher $\text{SO}_4^{2-}/\text{NO}_3^-$ ratio indicates more static sources, i.e., industry and energy production and lower $\text{SO}_4^{2-}/\text{NO}_3^-$ ratio indicates more mobile sources, i.e., traffic (Arimoto et al. 1996; Verma et al. 2010). Characteristic mass ratio of $\text{SO}_4^{2-}/\text{NO}_3^-$ was 2.0 over China and 1.4 over Asia in the year of 2000 (Zhang et al. 2013). The average $\text{SO}_4^{2-}/\text{NO}_3^-$ ratio in Bhola was 8.81 ± 4.08 which is significantly higher than in China. Verma et al. (2010) reported $\text{SO}_4^{2-}/\text{NO}_3^-$ ratio 6.93 and 13.6 for industrial emission in India during summer and fall. Although, a higher $\text{SO}_4^{2-}/\text{NO}_3^-$ ratio in Bhola indicates a static emission source, e.g., coal power stations in India. It must be, however, considered that this ratio can be as well influenced by the shift of particulate nitrate to gaseous nitric acid due to high ambient temperature during the sampling days.

Correlation among different chemical species determined from $\text{PM}_{2.5}$ is presented in Table 2, to consider association and the likely sources of chemical species. Among anions, NO_3^- showed strong correlation with nssK^+ , nssCa^{2+} , OC, EC, and mannitol indicating its mixed origin from sea, soil, and

Table 1 Minimum, maximum, and average concentrations ($\pm$ indicating standard deviation) of water-soluble ions, carbonaceous species, arabinol, and mannitol in $\text{PM}_{2.5}$ of Bhola Island. Levoglucosan, mannosan, and galactosan were below the detection limit. Total no. of samples analyzed = 9. * $\text{POM} = (1.6 \times \text{OC})$

	$\text{PM}_{2.5}$	OC	EC	OC/EC	POM*	Na^+	NH_4^+	nss K^+	nss Ca^{2+}	nss Mg^{2+}	Cl^-	NO_2^-	NO_3^-	nss SO_4^{2-}	Arabinol	Mannitol
Minimum	4.0	1.88	0.39	2.60	3.01	0.16	0.05	0.08	0.04	<LOD	0.02	0.04	0.03	0.77	0.04	0.04
Maximum	21.0	9.27	3.59	4.80	14.8	0.68	1.41	0.65	1.24	0.16	0.66	0.46	0.63	7.38	0.34	0.29
Average $\pm$ SD	15.0 ± 6.89	4.28 ± 2.48	1.29 ± 1.09	3.70 ± 0.88	6.85 ± 3.97	0.34 ± 0.20	0.34 ± 0.42	0.21 ± 0.19	0.32 ± 0.46	0.03 ± 0.05	0.17 ± 0.22	0.23 ± 0.14	0.23 ± 0.21	2.00 ± 2.06	0.11 ± 0.11	0.14 ± 0.09

All the units are in $\mu\text{g m}^{-3}$

Table 2 Correlation among different chemical species present in PM_{2.5} of Bhola Island, Bangladesh

	Na ⁺	NH ₄ ⁺	nssK ⁺	nssMg ²⁺	nssCa ²⁺	Cl [−]	NO ₂ [−]	NO ₃ [−]	nssSO ₄ ^{2−}	OC	EC	Arabitol	Mannitol
Na ⁺	1												
NH ₄ ⁺	0.37	1											
nssK ⁺	0.70	0.87	1										
nssMg ²⁺	0.83	−0.04	0.44	1									
nssCa ²⁺	0.90	0.44	0.78	0.89	1								
Cl [−]	0.77	−0.24	0.21	0.99	0.64	1							
NO ₂ [−]	−0.24	0.31	0.26	−0.18	0.00	−0.33	1						
NO ₃ [−]	0.86	0.61	0.91	0.74	0.93	0.54	0.24	1					
nssSO ₄ ^{2−}	0.98	0.40	0.76	0.89	0.99	0.83	−0.10	0.93	1				
OC	0.65	0.72	0.88	0.51	0.80	0.24	0.42	0.89	0.79	1			
EC	0.66	0.85	0.99	0.44	0.78	0.18	0.40	0.92	0.77	0.97	1		
Arbitol	0.01	0.36	0.14	−0.14	0.08	−0.18	0.25	0.13	0.08	0.49	0.89	1	
Mannitol	0.6	0.85	0.82	0.75	0.74	0.69	0.58	0.87	0.76	0.92	0.95	0.58	1

Bold significant at $p < 0.01$, two tailed

anthropogenic sources. NssSO₄^{2−} showed strong correlation with nssMg²⁺, nssCa²⁺, and NO₃[−] also indicating its mixed origin. NO₃[−] and nssSO₄^{2−} both showed strong correlation with Na⁺, which indicates long-range transport of aerosols over the sea. NssSO₄^{2−} and NO₃[−] could be generated from secondary aerosol particles formation of SO_x or NO_x gases. Among other anions, Cl[−] only showed strong correlation with Mg²⁺, which is typical for the sea water. Unexpectedly, relatively less strong correlation (0.77) was found between Cl[−] and Na⁺, which may be due to the Cl[−] depletion or there are other sources of Na⁺ (e.g., soil, biomass burning, or secondary aerosols) (Ooki et al. 2002). Among cations, nssCa²⁺ and nssMg²⁺ showed strong correlation with each other indicating their potential common sources such as soil or mineral dust. Strong correlation of Na⁺ with nssCa²⁺ indicates that soil/mineral dust contributes to Na⁺ in aerosol in addition to the sea salt, which is also evident from weak correlation between Na⁺ and Cl[−].

Enrichment factor of major ions

In order to separate and identify the main source of the determined water-soluble ions, we have calculated the enrichment factors (EFs). This calculation is based on the elemental ratio found between ions/elements collected in the aerosol, as compared to a similar ratio for a reference material (Huang et al. 2010). An EF value much less than or much higher than 1 is considered to be diluted or enriched relative to the reference source (Cao et al. 2009). EFs were calculated by using following equations using Na and Ca as reference materials for sea and crust, respectively (Nishikawa et al. 1991; Cao et al. 2009):

$$EF_{\text{sea water}} = (X/Na)_{\text{aerosol}} / (X/Na)_{\text{sea}} \quad (4)$$

$$EF_{\text{crust}} = (X/Ca)_{\text{aerosol}} / (X/Ca)_{\text{soil}} \quad (5)$$

where, X is the concentration of the species of interest. (X/Na)_{sea} is the ratio from seawater composition (Kidron and Starinsky 2012; Xiao 2016) and (X/Ca)_{crust} is the ratio from crustal composition (Taylor 1964).

Based on the value of EF, they may be subdivided into the following classes (Azari et al. 2010):

(i) EF > 10, slightly enriched; (ii) 10 < EF < 1000, enriched; (iii) EF > 1000, highly enriched (Table 3).

K⁺ and Mg²⁺ are slightly enriched compared to both sea and crust, which indicates their mixed origin from those sources. Although fine particles of K⁺ could also result from biomass burning, it is difficult to distinguish them from other sources, as levoglucosan was found below detection limit. Cl[−] mainly came from sea as it is slightly enriched compared to sea, but enriched compared to crust. NO₃[−] and SO₄^{2−} are enriched and highly enriched compared to both sea and crust. So their contributions from anthropogenic sources must be stronger than other sources.

Table 3 Enrichment factors of water-soluble ions determined from PM_{2.5} in Bhola Island, Bangladesh

	K ⁺	Mg ²⁺	Cl [−]	NO ₃ [−]	SO ₄ ^{2−}
EF _{sea}	2.89	2.51	0.42	33,965	92.9
EF _{crust}	1.32	0.36	162.2	328.3	328.9

*The EFs of Na and Ca were not calculated as we considered them as pure sea and crustal-derived species

Saccharides

Levoglucosan, mannosan, and galactosan were below detection limit in all collected samples. One important thing that should be noted here is that levoglucosan is a potential biomass marker in aerosol and ubiquitous in the atmosphere at trace level to higher concentration (Villalobos et al. 2015; Hu et al. 2013). We might not detect it due to loss pathway like photo-oxidation (Hennigan et al. 2010). Levoglucosan may be present in small concentration ($<0.02 \mu\text{g m}^{-3}$), which indicates that biomass burning had only very little or no influence on $\text{PM}_{2.5}$ during the sampling days. On the other hand, sugar alcohols: arabitol and mannitol were detected in every sample. The concentration of arabitol was in between 0.04 and $0.34 \mu\text{g m}^{-3}$ with an average $0.11 \mu\text{g m}^{-3}$, and the concentration of mannitol was in between 0.04 and $0.29 \mu\text{g m}^{-3}$ with an average $0.14 \mu\text{g m}^{-3}$. Presence of arabitol and mannitol is a confirmation for biological activity in the particles. The concentrations may vary from region to region depending on vegetation characteristics, season, meteorological conditions, geographical location, and agricultural activities (Zhang et al. 2010). The average concentrations of arabitol and mannitol in Bhola are comparable to studies conducted in south China (Zhang et al. 2010), Brazil (Graham et al. 2003; Decesari et al. 2006), and Germany (Kourichev et al. 2008). In contrast, the average concentration of those species in Bhola is significantly lower than wintertime aerosol of rural India (Nirmalkar et al. 2015). The total concentration of arabitol and mannitol was 1.67% of $\text{PM}_{2.5}$ mass. The potential sources of arabitol and mannitol in aerosol are fungal spores, biomass burning, and microbial-active soil re-suspension (Bauer et al. 2008; Mims and Mims 2004; Claeys et al. 2010). In the correlation analysis, arabitol and mannitol do not show significant correlation with each other indicating that they were either not from common sources or have a different atmospheric stability. Arabitol showed significant correlation with EC only. Mannitol showed strong correlation with EC, OC, NH_4^+ , and NO_3^- . The correlation analysis suggests that mannitol is coming from mixed sources, most probably was transported over the sampling region together with aerosol enriched by combustion processes and biogenic emissions. Small biomass burning event might have contributed in the generation of arabitol and mannitol, as both showed significant correlation with EC what was observed also in the case of biomass burning events (Popovicheva et al. 2014).

Carbonaceous species

The average concentration of OC and EC in $\text{PM}_{2.5}$ of Bhola were 4.3 ± 2.5 and $1.3 \pm 1.1 \mu\text{g m}^{-3}$, respectively. OC was the main contributor of total carbon (TC) and accounted for about 28% of $\text{PM}_{2.5}$. Sum of EC and organic matter (OM) was used to calculate total carbonaceous aerosol using a scaling factor

from the literatures ($\text{TCA} = \text{OM} + \text{EC} = 1.6 \times \text{OC} + \text{EC}$) (Turpin and Huntzicker 1995; Cao et al. 2003; Gu et al. 2010; Pachauri et al. 2013). The average concentration of TCA was $8.0 \pm 5.0 \mu\text{g m}^{-3}$. The calculated TCA account for the 57.1% of $\text{PM}_{2.5}$ mass and indicates that fine particles were containing predominantly carbonaceous aerosol. The mass ratio of OC to EC can be used to interpret the contribution of fresh combustion emissions and transformation characteristics of carbonaceous aerosol (Turpin and Huntzicker 1995; Rajput et al. 2014a). OC/EC is influenced by emission sources, secondary organic aerosol (SOA) formation, and different removal rates by deposition (Cachier et al. 1996; Satsangi et al. 2012). The OC/EC ratio in Bhola was 3.70 ± 0.88 . The OC/EC was greater than 2, indicating secondary organic aerosol formation in addition to primary emission sources (Pachauri et al. 2013; Satsangi et al. 2012). The secondary organic carbon (SOC) fraction in aerosol was estimated based on the minimal OC/EC method (Turpin and Huntzicker 1995; Tang et al. 2016).

$$\text{SOC} = \text{OC}_{\text{total}} - \text{EC} \times (\text{OC/EC})_{\text{min}} \quad (6)$$

where, OC_{total} is the total OC and $(\text{OC/EC})_{\text{min}}$ is the minimum ratio of OC to EC concentration monitored. This method was reported to be robust for larger data sets; thus, it might be burdened with somewhat higher uncertainty in case of a small sample set in this study.

The concentration of SOC was $1.48 \pm 1.65 \mu\text{g m}^{-3}$, which is 25.7% of OC. Ram and Sarin (2010) observed ~30% of SOC in OC from Northern India, which gives a good hope for the performance of used model also in our data set.

Strong correlation was observed between OC and EC (0.97). The higher OC/EC ratios (or higher OC/TC ratio) indicate biomass burning or secondary aerosol (aged biomass burning and biogenic aerosol) as a major source; whereas lower ratios are characteristics of emissions from fossil fuel (Ram and Sarin 2010). Several studies have reported OC/EC ratio for various emission sources (Pachauri et al. 2013; Mandal et al. 2014), which includes vehicular exhaust, 2.5–5.0 (Schauer et al. 2002); coal smoke, 2.5–10.5 (Chen et al. 2006); kitchen emissions, 4.3–7.7 (See and Balasubramaniam 2008); biomass burning, 3.8–13.2 (Zhang et al. 2007); and Traffic, 4.4 (Pachauri et al. 2013; Mandal et al. 2014). Rajput et al. (2014a) reported OC/EC in between 2.0 and 6.5 during post-harvesting (April–May) season in Patiala, India, and inferred the source as biomass burning. The OC/EC ratio of the present study (2.6–4.8) matches with mixed source-derived aerosol of rural Northern India (Ram and Sarin 2010), where OC/EC ratio was 2.1–4.0. Two major sources of EC and OC are fossil fuels and biomass burning. K^+ is generally used as an index of biomass burning due to its release during combustion processes. In contrast, combustion of fossil-fuel emission produces very little K^+ (Andreae 1983;

Satsangi et al. 2012). In this study, nssK^+ showed very good correlation with EC (0.99) and OC (0.88) indicating their common origin. The relatively high K^+/EC ratios have been reported earlier for biomass burning (range 0.21–0.46) and low ratios for fossil-fuel emissions (range 0.025–0.09) (Ram and Sarin 2010; Pachauri et al. 2013). The K^+/EC ratios were between 0.16 and 0.25 with an average of 0.20, which is greater than the ratio of fossil-fuel emission and close to the lower limit of biomass burning. K^+/OC and nssK^+/OC were also calculated, both values were between 0.02 and 0.08. Pachauri et al. (2013) reported K^+/OC ratio 0.04–0.13 for agricultural waste burning in IGP. Whereas Rajput et al. (2014a) reported nssK^+/OC ratio 0.01–0.13 for fossil/biofuel emission in the IGP region.

The ratio between $\text{nssSO}_4^{2-}/\text{OC}$ in the present study was 0.17–0.80 with an average 0.47 ± 0.21 . Besides, the ratio between $\text{nssSO}_4^{2-}/\text{EC}$ was between 0.81 and 2.41 with an average of 1.70 ± 0.68 . According to Rajput et al. (2014b) in the IGP, the $\text{nssSO}_4^{2-}/\text{OC}$ ratio is about 0.2–0.5, whereas the $\text{nssSO}_4^{2-}/\text{EC}$ ratio is 2.0–4.0. $\text{nssSO}_4^{2-}/\text{EC}$ ratio could serve as an important proxy for identifying the relative dominance of contribution from fossil-fuel combustion in relation to biomass burning emissions (Srinivas and Sarin 2014). Srinivas and Sarin (2014) mentioned that the $\text{nssSO}_4^{2-}/\text{EC}$ ratio in fine mode aerosols from the biomass-related emission can be 0.70 ± 0.10 (Andreae and Merlet 2001), 0.28 ± 0.13 (Ferek et al. 1998), and 0.32 ± 0.20 (Mkoma et al. 2013). $\text{nssSO}_4^{2-}/\text{EC}$ ratio of the present study and IGP (Rajput et al. 2014b; Srinivas and Sarin 2014) is higher than those literature values.

It is widely known that SO_4^{2-} is predominantly a secondary aerosol component forming through aqueous-mediated gas-to-particulate phase conversion of primarily emitted SO_2 (Rajput et al. 2014b). NH_4^+ and SO_4^{2-} are two of major ions from biomass burning aerosol (Sillapapiromsuk et al. 2013; Dambruoso et al. 2014) which may exist as ammonium sulfate after gas-to-particulate phase conversion. Although, correlation between nssSO_4^{2-} and NH_4^+ were very weak (0.40). However, OC can be produced directly from its emission source (primary origin) and it can also undergo secondary formation, such as SO_4^{2-} (Rajput et al. 2014b). It has been reported in recent literatures that aerosol acidity derived from nss-SO_4^{2-} favors secondary organic aerosol formation from biogenic emissions (Surratt et al. 2007, 2008; Rajput et al. 2014b). An $\text{NH}_4^+/\text{SO}_4^{2-}$ equivalent ratio less than 1 is in generally considered to be an index for aerosol acidity from nss-SO_4^{2-} (Rengarajan et al. 2011; Rajput et al. 2014b). In our present study, $\text{NH}_4^+/\text{SO}_4^{2-}$ equivalent ratio was in between 0.06 and 0.55 with an average 0.30 ± 0.13 indicating favorable condition for the formation of SO_4^{2-} containing SOA. The formation of organosulfates has been reported in an urban location of Western India; where $\text{NH}_4^+/\text{SO}_4^{2-}$ equivalent ratio varying from 0.60 to 0.80 (Rengarajan et al. 2011; Rajput et al. 2014b). This is also in line if the ion balance is considered, as

several days show a cation excess and it could be true that organic ions like oxalate, formed as secondary products are missing. Biomass burning can serve as a potential source of NH_4^+ (Andreae and Merlet 2001). But NH_4^+ showed comparatively weak correlation with EC (0.85) and OC (0.72). The potential source of NH_4^+ can be secondary aerosol formation by NH_3 gas emitted from agricultural activities.

In conclusion, different ratio analyses suggested carbonaceous species came from mixed sources. Although, some ratios (K^+/EC , $\text{nssSO}_4^{2-}/\text{EC}$) are much higher than the fossil-fuel burning ratio and some are (K^+/OC , nssK^+/OC) very close to biomass burning ratio. In addition to that, a strong correlation was found between nssK^+ and carbonaceous components (OC, $R^2 = 0.88$; EC, $R^2 = 0.99$). Those give a hint that there may be influence (as levoglucosan was below detection limit) from biomass burning during this sampling period but rather not in the nearest vicinity. As IGP is called “bread basket” of South Asia, it can be expected that firing of agricultural waste might occur on event base. Furthermore, there may be influence of long-range transport as well.

Backward air mass trajectories analysis

Frequency of the backward air mass trajectories during monsoon showed that air masses originated from the Arabian Sea and passed through the southern part of India, Bay of Bengal before converging at Bhola island (Fig. 2a). Part of continent which is crisscrossed by the air masses could be the major source of long-range pollutant during May to August sampling. On the other hand, during winter season, air masses originated from the northwest part of Indian subcontinent and traveled through the IGP region (over land most of the time) before converging at Bhola island (Fig. 2b). When air masses were originating from the IGP region (over land), the concentration was higher than the air masses originating from the Bay of Bengal. For example, the concentrations of OC, ($\text{SO}_4^{2-} + \text{NO}_3^- + \text{NH}_4^+$), ($\text{Ca}^{2+} + \text{K}^+$), and (arabitol+mannitol) on April 27, 2013 (Fig. 2c) were 2.15, 1.09, 1.35, and 4.73 times higher than the concentrations of August 4, 2017 (Fig. 2d), respectively. The rest of the sampling days, the backward air mass trajectories are similar to Fig. 2d.

Conclusion

The chemical characterization of $\text{PM}_{2.5}$ aerosols collected from rural Bhola Island, Bangladesh, during pre-monsoon and monsoon season has been presented. The major chemical species present in $\text{PM}_{2.5}$ was carbonaceous materials (OC, EC) followed by nssSO_4^{2-} , Na^+ , and NH_4^+ . Although Bhola is an island of the Bay of Bengal, but percent contribution of sea salt aerosol was lower (17.2%). Chloride depletion was high with an average 78.2%. SO_4^{2-} and NO_3^- were enriched/

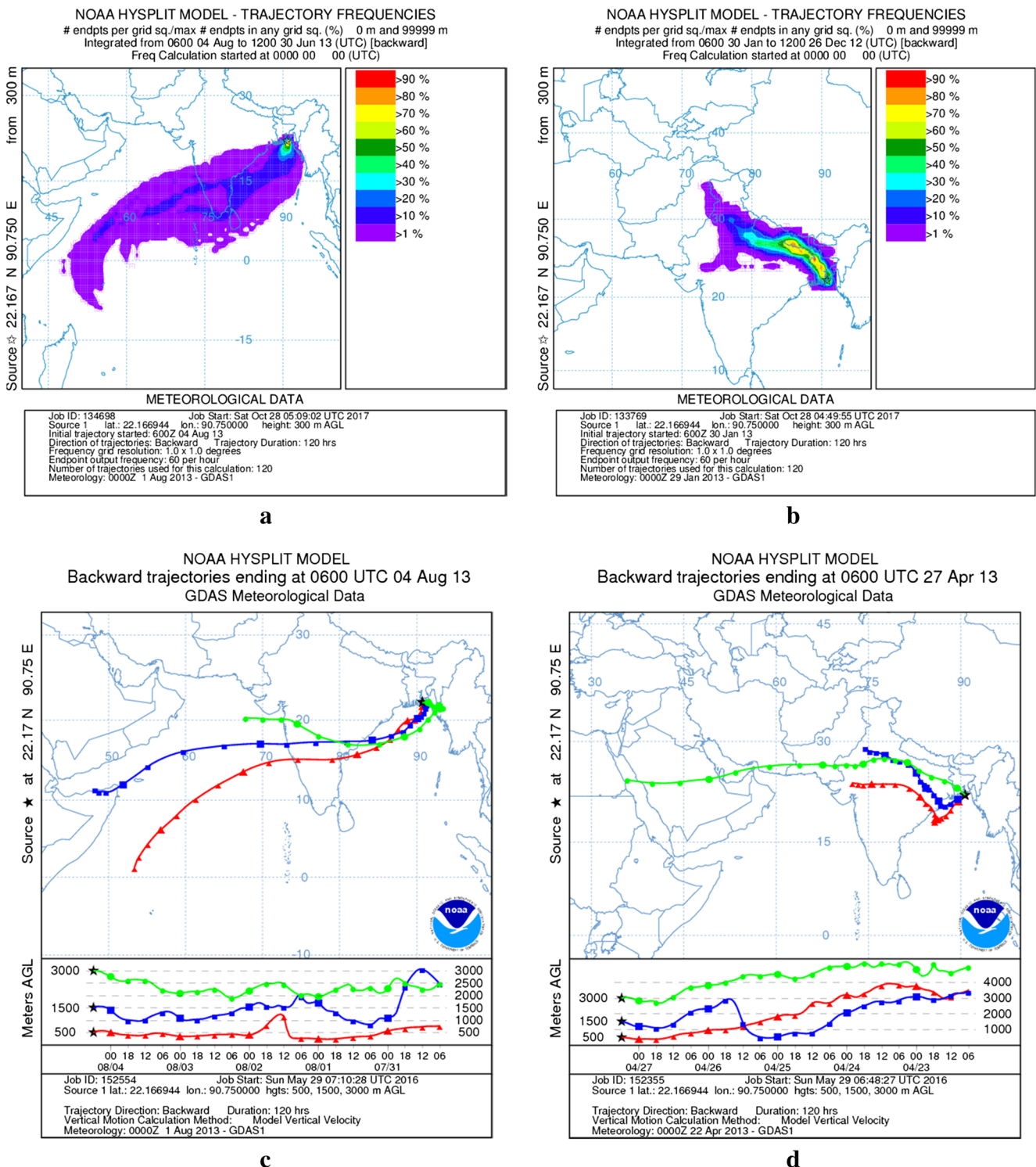


Fig. 2 Top panels are the frequency of the backward air mass trajectories (NOAA, HYSPLIT) during monsoon (a) and winter (b). Bottom panels are the selected two-days (c August 4, 2017; d April 27, 2013) backward air mass trajectories during sampling in Bhola Island, Bangladesh

highly enriched in aerosol compared to sea and soil indicating their main source as anthropogenic. $\text{NH}_4^+/\text{SO}_4^{2-}$ ratio was found favorable for the formation of SO_4^{2-} containing SOA. Higher OC/EC ratio confirmed as well the secondary organic carbon formation. Ratio analysis suggested that carbonaceous

species came from mixed sources including biomass burning, but only to a low extent. Arabitol and mannitol have different source profiles but both are influenced by biomass burning. Air mass trajectories analysis showed that concentration of anthropogenic and crustal species are higher on April, when

air originated and travels most of the time through land. The results from this study provide the regional background characteristics of fine aerosol particles in Bangladesh.

Acknowledgements Authors acknowledge the financial support from the Office of the Naval Research Global (ONRG), USA, for construction of the Bhola observatory. Thanks to the sampling crews of Bhola observatory.

References

- Andreae MO (1983) Soot carbon and excess fine potassium: long-range transport of combustion-derived aerosols. *Science* 220(4602):1148–1151. <https://doi.org/10.1126/science.220.4602.1148>
- Andreae MO, Merlet P (2001) Emission of trace gases and aerosols from biomass burning. *Glob Biogeochem Cycles* 15(4):955–966. <https://doi.org/10.1029/2000GB001382>
- Arfelli G, Sartini E (2014) Characterization of brewpub beer carbohydrates using high performance anion exchange chromatography coupled with pulsed amperometric detection. *Food Chem* 142:152–158. <https://doi.org/10.1016/j.foodchem.2013.07.008>
- Arimoto R, Duce RA, Savoie DL, Prospero JM, Talbot RW, Cullen JD, Tomza U, Lewis NF, Ray BJ (1996) Relationships among aerosol constituents from Asia and the North Pacific during PEM-West A. *J Geophys Res* 101(D1):2011–2023. <https://doi.org/10.1029/95JD01071>
- Azari C, Mabrouk C, Abida H, Medhioub K (2010) Water soluble component in PM₁₀ aerosols over an urban and a suburban site in the city of Sfax (Tunisia). *Atmosfera* 23:197–211
- Bates TS, Quinn PK, Coffman D, James E, Miller T, Covert D, Wiedensohler A, Leinert S, Nowak A, Neus C (2001) Regional physical and chemical properties of the marine boundary layer aerosol across the Atlantic during Aerosols99: an overview. *J Geophys Res* 106(D18):20767–20782. <https://doi.org/10.1029/2000JD900578>
- Bauer H, Schueller E, Weinke G, Berger A, Hitzemberger R, Marr IL, Puxbaum H (2008) Significant contributions of fungal spores to the organic carbon and to the aerosol mass balance of the urban atmospheric aerosol. *Atmos Environ* 42(3):588–593. <https://doi.org/10.1016/j.atmosenv.2007.10.013>
- Begum B, Kim E, Biswas SK, Hopke PK (2004) Investigation of sources of atmospheric aerosol at urban and semi-urban areas in Bangladesh. *Atmospheric Environment* 38(19):3025–3038
- Begum BA, Biswas SK, Markwitz A, Hopke PK (2010) Identification of sources of fine and coarse particulate matter in Dhaka, Bangladesh. *Aerosol Air Qual Res* 10:345–353
- Begum BA, Hossain A, Nahar N, Markwitz A, Hopke PK (2012) Organic and black carbon in PM_{2.5} at an urban site at Dhaka, Bangladesh. *Aerosol Air Qual Res* 12:1062–1072
- Birch ME, Cary RA (1996) Elemental carbon based for monitoring occupational exposures to particulate diesel exhaust. *Aerosol Sci Technol* 25(3):221–241. <https://doi.org/10.1080/02786829608965393>
- Cachier H, Liousse C, Pertuisol MH, Gaudichet A, Echalar F, Lacaux JP (1996) African fire particulate emissions and atmospheric influence. In: Levine EJS (ed) *Biomass burning and global change*. MIT Press, London, pp 428–440
- Cao JJ, Lee SC, Ho KF, Zhang XY, Zou SC, Fung K, Chow JC, Watson JG (2003) Characteristics of carbonaceous aerosol in Pearl River Delta Region, China during 2001 winter period. *Atmos Environ* 37(11):1451–1460. [https://doi.org/10.1016/S1352-2310\(02\)01002-6](https://doi.org/10.1016/S1352-2310(02)01002-6)
- Cao YZ, Wang S, Zhang G, Luo J, Lu S (2009) Chemical characteristics of wet precipitation at an urban site of Guangzhou, South China. *Atmos Res* 94(3):462–469. <https://doi.org/10.1016/j.atmosres.2009.07.004>
- Caseiro A, Bauer H, Schmidl C, Pio CA, Puxbaum H (2009) Wood burning impact on PM₁₀ in three Austrian regions. *Atmos Environ* 43(13):2186–2195. <https://doi.org/10.1016/j.atmosenv.2009.01.012>
- Chattopadhyay S, Gupta S, Saha RN (2010) Spatial and temporal variation of urban air quality: a GIS approach. *J Environ Prot* 1(03):264–277. <https://doi.org/10.4236/jep.2010.13032>
- Chen Y, Zhi G, Feng Y, Fu J, Feng J, Shen G, Simoneit BRT (2006) Measurement of emission factors for primary carbonaceous particles from residential raw coal combustion in China. *Geophys Res Lett* 33(2). <https://doi.org/10.1029/2006GL026966>
- Claeys M, Kourtev I, Pashynska V, Vermeylen R, Wang W, Cafmeyer J, Chi X, Artaxo P, Andreae MO, Maenhaut W (2010) Polar organic marker compounds in atmospheric aerosols during the LBA-SMOCC 2002 biomass burning experiment in Rondonia, Brazil: sources and source processes, time series, diel variations and size distributions. *Atmos Chem Phys* 10(19):9319–9331. <https://doi.org/10.5194/acp-10-9319-2010>
- Dambruoso P, de Gennaro G, Di Gilio A, Palmisani J, Tutino M (2014) The impact of infield biomass burning on PM levels and its chemical composition. *Environ Sci Pollut Res* 21(23):13175–13185. <https://doi.org/10.1007/s11356-013-2384-4>
- Decesari S, Fuzzi S, Facchini MC, Mircea M, Emblico L, Cavalli F, Maenhaut W, Chi X, Schkolnik G, Favkovich A, Rudich Y, Claeys M, Pashynska V, Kourtev I, Vermeylen R, Hoffer A, Andreae MO, Tagliavini E, Moretti F, Artaxo P (2006) Characterization of the organic composition of aerosols from Rondônia, Brazil, during the LBA-SMOCC 2002 experiment and its representation through model compounds. *Atmos Chem Phys* 6(2):375–402. <https://doi.org/10.5194/acp-6-375-2006>
- Ferek RJ, Reid JS, Hobbs PV, Blake DR, Liousse C (1998) Emission factors of hydrocarbons, halocarbons, trace gases and particles from biomass burning in Brazil. *J Geophys Res* 103(D24):32107–32118. <https://doi.org/10.1029/98JD00692>
- Graham B, Guyon P, Taylor PE, Artaxo P, Maenhaut W, Glovsky MM, Flagan RC, Andreae MO (2003) Organic compounds present in the natural Amazonian aerosol: characterization by gas chromatography-mass spectrometry. *J Geophys Res* 108:4766
- Gu J, Bai Z, Liu A, Wu L, Xie Y, Li W, Dong H, Zhang X (2010) Characterization of atmospheric organic carbon of PM_{2.5} and PM₁₀ at Tianjin, China. *Aerosol Air Qual Res* 10:167–176
- Hennigan CJ, Sullivan AP, Collett JL, Robinson AL (2010) Levoglucosan stability in biomass burning particles exposed to hydroxyl radicals. *Geophys Res Lett* 37(9). <https://doi.org/10.1029/2010GL043088>
- Hu QH, Xie ZQ, Wang X, Kang H, Zhang P (2013) Levoglucosan indicates high levels of biomass burning aerosols over oceans from the Arctic to Antarctic. *Sci Rep* 3:3119
- Huang XF, Li X, He LY, Feng N, Hu M, Niu YW, Zeng LW (2010) 5-year study of rainwater chemistry in a coastal mega-city in South China. *Atmos Res* 97:185–193
- Inuma Y, Englin G, Puxbaum H, Herrmann H (2009) A highly resolved anion-exchange chromatographic method for determination of saccharidic tracers for biomass combustion and primary bio-particles in atmospheric aerosol. *Atmos Environ* 43(6):1367–1371. <https://doi.org/10.1016/j.atmosenv.2008.11.020>
- Jusino-Atresino R, Anderson J, Gao Y (2016) Ionic and elemental composition of PM_{2.5} aerosols over the Caribbean Sea in the Tropical Atlantic. *J Atmos Chem* 73(4):427–457. <https://doi.org/10.1007/s10874-016-9337-5>
- Kidron GJ, Starinsky A (2012) Chemical composition of dew and rain in an extreme desert (Negev): cobbles serve as sink for nutrients. *J Hydrol* 420–421:284–291

- Kourtchev I, WamkeJ MW, Hoffmann T, Claeys M (2008) Polar organic marker compounds in PM_{2.5} aerosol from a mixed forest site in western Germany. *Chemosphere* 73(8):1308–1314. <https://doi.org/10.1016/j.chemosphere.2008.07.011>
- Mandal P, Sarkar R, Mandal A, Saud T (2014) Seasonal variation and sources of aerosol pollution in Delhi, India. *Environ Chem Lett* 12(4):529–534. <https://doi.org/10.1007/s10311-014-0479-x>
- Mims SA, Mims IIFM (2004) Fungal spores are transported long distances in smoke from biomass fires. *Atmos Environ* 38(5):651–655. <https://doi.org/10.1016/j.atmosenv.2003.10.043>
- Mkoma SL, Kawamura K, Fu P (2013) Contributions of biomass/biofuel burning to organic aerosols and particulate matter in Tanzania, East Africa, based on analyses of ionic species, organic and elemental carbon, levoglucosan and mannosan. *Atmos Chem Phys* 13(20):10325–10338. <https://doi.org/10.5194/acp-13-10325-2013>
- Mondol MN, Khaled M, Chamon AS, Ullah SM (2014) Trace metal concentration in atmospheric aerosols in some city areas of Bangladesh. *Bangladesh J Sci Ind Res* 49:263–270
- Nirmalkar J, Deb MK, Tsai YI, Deshmukh DK (2015) Arabitol and mannitol as tracer for fungal contribution to size-differentiated particulate matter of rural atmospheric aerosols. *Int J Environ Sci Dev* 6(6):460–463. <https://doi.org/10.7763/IJESD.2015.V6.637>
- Nishikawa M, Kanamori S, Kanamori N, Mizoguchi T (1991) Kosa aerosol as eolian carrier of anthropogenic material. *Sci Total Environ* 107:13–27. [https://doi.org/10.1016/0048-9697\(91\)90247-C](https://doi.org/10.1016/0048-9697(91)90247-C)
- Ooki A, Uematsu M, Miura K, Nakae S (2002) Sources of sodium in atmospheric fine particles. *Atmos Environ* 36(27):4367–4374. [https://doi.org/10.1016/S1352-2310\(02\)00341-2](https://doi.org/10.1016/S1352-2310(02)00341-2)
- Pachauri T, Satsangi A, Singla V, Lakhani A, Kumari KM (2013) Characteristics and sources of carbonaceous aerosols in PM_{2.5} during wintertime in Agra, India. *Aerosol Air Qual Res* 13:977–991
- Pakkanen TA (1996) Study of formation of coarse particle nitrate aerosol. *Atmos Environ* 30(14):2475–2482
- Pawar H, Garg S, Kumar V, Sachan H, Arya R, Sarkar C, Chandra BP, Sinha B (2015) Quantifying the contribution of long-range transport to particulate matter (PM) mass loadings at a suburban site in the north-western Indo-Gangetic Plain (NW-IGP). *Atmos Chem Phys* 15(16):9501–9520. <https://doi.org/10.5194/acp-15-9501-2015>
- Popovicheva OB, Kistler M, Kireeva E, Persiantseva N, Timofeev M, Kopeikin V, Kasper-Giebl A (2014) Physicochemical characterization of smoke aerosol during large-scale wildfires: extreme event of August 2010 in Moscow. *Atmos Environ* 96:405–414. <https://doi.org/10.1016/j.atmosenv.2014.03.026>
- Rajput P, Sarin MM, Sharma D, Singh D (2014a) Characteristics and emission budget of carbonaceous species from post-harvest agricultural-waste burning in source region of the Indo-Gangetic Plain. *Tellus B* 66(1). <https://doi.org/10.3402/tellusb.v66.21026>
- Rajput P, Sarin MM, Sharma D, Singh D (2014b) Organic aerosols and inorganic species from post-harvest agricultural-waste burning emissions over northern India: impact on mass absorption efficiency of elemental carbon. *Environ Sci Process Impacts* 16:2371–2379
- Ram K, Sarin MM (2010) Spatio-temporal variability in atmospheric abundances of EC, OC and WSOC over Northern India. *J Aerosol Sci* 41(1):88–98. <https://doi.org/10.1016/j.jaerosci.2009.11.004>
- Rana M, Sulaiman M, Sivertsen N, Khan B, Nasreen S (2016) Trends in atmospheric particulate matter in Dhaka, Bangladesh, and the vicinity. *Environ Sci Pollut Res* 23(17):17393–17403. <https://doi.org/10.1007/s11356-016-6950-4>
- Rengarajan R, Sudheer AK, Sarin MM (2011) Aerosol acidity and secondary organic aerosol formation during wintertime over urban environment in Western India. *Atmos Environ* 45(11):1940–1945. <https://doi.org/10.1016/j.atmosenv.2011.01.026>
- Salam A, Bauer H, Kassim K, Ullah S, Puxbaum H (2003a) Aerosol chemical characteristics of a mega-city in Southeast Asia (Dhaka–Bangladesh). *Atmos Environ* 37(18):2517–2528. [https://doi.org/10.1016/S1352-2310\(03\)00135-3](https://doi.org/10.1016/S1352-2310(03)00135-3)
- Salam A, Bauer H, Kassim K, Ullah SM, Puxbaum H (2003b) Aerosol chemical characteristics of an island site in the Bay of Bengal (Bhola–Bangladesh). *J Environ Monit* 5(3):483–490. <https://doi.org/10.1039/b212521h>
- Salam A, Hossain T, Siddique TMNA, Alam AMS (2008) Characteristics of atmospheric trace gases, particulate matter, and heavy metal pollution in Dhaka, Bangladesh. *Air Qual Atmos Health* 1(2):101–109. <https://doi.org/10.1007/s11869-008-0017-8>
- Salam A, Al-Mamoon H, Ullah MB, Ullah SM (2012) Measurement of the atmospheric aerosol particle size distribution in a highly polluted mega-city in Southeast Asia (Dhaka–Bangladesh). *Atmos Environ* 59:338–343. <https://doi.org/10.1016/j.atmosenv.2012.05.024>
- Satsangi A, Pachauri T, Singla V, Lakhani A, Kumari KM (2012) Organic and elemental carbon aerosols at a suburban site. *Atmos Res* 113:13–21. <https://doi.org/10.1016/j.atmosres.2012.04.012>
- Saud T, Singh DP, Mandal TK, Ranu G, Pathak H, Saxena M, Sharma SK, Gautam R, Mukherje A, Bhatnagar RP (2011) Spatial distribution of biomass consumption as energy in rural areas of the Indo-Gangetic Plain. *Biomass Bioenergy* 35(2):932–941. <https://doi.org/10.1016/j.biombioe.2010.11.001>
- Schauer JJ, Kleeman MJ, Cass GR, Simoneit BRT (2002) Measurement of emissions from air pollution sources. 5. C1–C32 organic compounds from gasoline-powered motor vehicles. *Environ Sci Technol* 36(6):1169–1180. <https://doi.org/10.1021/es0108077>
- See SW, Balasubramanian R (2008) Chemical characteristics of fine particles emitted from different gas cooking methods. *Atmos Environ* 42(39):8852–8862. <https://doi.org/10.1016/j.atmosenv.2008.09.011>
- Shahid I, Kistler M, Mukhtar A, Ghauri B, Ramirez-Santa Cruz C, Bauer H, Puxbaum H (2016) Chemical characterization and mass closure of PM₁₀ and PM_{2.5} at an urban site in Karachi–Pakistan. *Atmos Environ* 128:114–123. <https://doi.org/10.1016/j.atmosenv.2015.12.005>
- Shohel M, Simol HA, Reid E, Reid JS, Salam A (2017) Dew water chemical composition and source characterization in the IGP outflow location (coastal Bhola, Bangladesh). *Air Qual Atmos Health* 10(8):981–990. <https://doi.org/10.1007/s11869-017-0487-7>
- Sillapapiromsuk S, Chantara S, Tengjaroenkul U, Prasitwattanaseree S, Prapamontol T (2013) Determination of PM₁₀ and its ion composition emitted from biomass burning in the chamber for estimation of open burning emissions. *Chemosphere* 93(9):1912–1919. <https://doi.org/10.1016/j.chemosphere.2013.06.071>
- Sinha V, Kumar V, Sarkar C (2014) Chemical composition of pre-monsoon air in the Indo-Gangetic Plain measured using a new air quality facility and PTR-MS: high surface ozone and strong influence of biomass burning. *Atmos Chem Phys* 14(12):5921–5941. <https://doi.org/10.5194/acp-14-5921-2014>
- Snider G, Weagle CL, Murdymootoo KK, Ring A, Ritchie Y, Walsh A, Akoshile C, Anh NX, Brook J, Qonitan FD, Dong J, Griffith D, Holben B, Kahn R, Lagrosas N, Lestari P, Ma Z, Misra A, Quel EJ, Salam A, Schichtel B, Segev L, Tripathi SN, Wang C, Yu C, Zhang Q, Zhang Y, Brauer M, Cohen A, Gibson MD, Liu Y, Martins JV, Rudich Y, Martin RV (2016) Variation in global chemical composition of PM_{2.5}: emerging results from SPARTAN. *Atmos Chem Phys* 16(15):9629–9653. <https://doi.org/10.5194/acp-16-9629-2016>
- Srinivas B, Sarin MM (2014) PM_{2.5}, EC and OC in atmospheric outflow from the Indo-Gangetic Plain: temporal variability and aerosol organic carbon-to-organic mass conversion factor. *Sci Total Environ* 487:196–205. <https://doi.org/10.1016/j.scitotenv.2014.04.002>
- Styszko K, Samek L, Szramowiat K, Korzeniewska A, Kubisty K, Rakoczy-Lelek R, Giebl AK (2017) Oxidative potential of PM₁₀ and PM_{2.5} collected at high air pollution site related to chemical composition: Krakow case study. *Air Qual Atmos Health*:1–15. <https://doi.org/10.1007/s11869-017-0499-3>
- Surratt JD, Lewandowski M, Offenberg JH, Jaoui M, Kleindienst TE, Edney EO, Seinfeld JH (2007) Effect of acidity on secondary

- organic aerosol from isoprene. *Environ Sci Technol* 41(15):5363–5369. <https://doi.org/10.1021/es0704176>
- Surratt JD, Gómez-González Y, AWH C, Vermeulen R, Shahgholi M, Kleindienst TE, Edney EO, Offenberg JH, Lewandowski M, Jaoui M, Maenhaut W, Claeys M, Flagan RC, Seinfeld JH (2008) Organosulfate formation in biogenic secondary organic aerosol. *J Phys Chem A* 112(36):8345–8378. <https://doi.org/10.1021/jp802310p>
- Taneja T, Pal DB, Joshi PK, Aggarwal PK, Tyagi NK (2014) Farmers preferences for climate-smart agriculture: an assessment in the Indo-Gangetic Plain. IFPRI discussion paper 01337, May 2014
- Tang M, Ji DS, Gao WK, ZW Y, Chen K, Cao W (2016) Characteristics of air quality in Tianjin during the Spring Festival period of 2015. *Atmos Ocean Sci Lett* 9:15–21
- Taylor SR (1964) Abundance of chemical elements in the continental crust: a new table. *Geochim Cosmochim Acta* 28(8):1273–1285. [https://doi.org/10.1016/0016-7037\(64\)90129-2](https://doi.org/10.1016/0016-7037(64)90129-2)
- Tiwari S, Dumka UC, Hopke PK, Tunved P, Srivastava AK, Bisht DS, Chakrabarty RK (2016) Atmospheric heating due to black carbon aerosol during the summer monsoon period over Ballia: a rural environment over Indo-Gangetic Plain. *Atmos Res* 178–179:393–400. <https://doi.org/10.1016/j.atmosres.2016.04.008>
- Turpin BJ, Huntzicker JJ (1995) Identification of secondary organic aerosol episodes and quantification of primary and secondary organic aerosol concentrations during SCAQS. *Atmos Environ* 29(23):3527–3544. [https://doi.org/10.1016/1352-2310\(94\)00276-Q](https://doi.org/10.1016/1352-2310(94)00276-Q)
- Verma SK, Deb MK, Suzuki Y, Tsai YI (2010) Ion chemistry and source identification of coarse and fine aerosols in an urban area of eastern central India. *Atmos Res* 95(1):65–76. <https://doi.org/10.1016/j.atmosres.2009.08.008>
- Villalobos AM, Amonov MO, Shafer MM, Devi JJ, Gupta T, Tripathi SN, Rana KS, Mckenzie M, Bergin MH, Schauer JJ (2015) Source apportionment of carbonaceous fine particulate matter (PM_{2.5}) in two contrasting cities across the Indo-Gangetic Plain. *Atmos Pollut Res* 6(3):398–405. <https://doi.org/10.5094/APR.2015.044>
- Virkkula A, Teinilä K, Hillamo R, Kerminen VM, Saarikoski S, Aurela M, Koponen IK, Kulmala M (2006) Chemical size distributions of boundary layer aerosol over the Atlantic Ocean and at an Antarctic site. *J Geophys Res* 111(D5):D05306
- Wahid A (2006) Influence of atmospheric pollutants on agriculture in developing countries: a case study with three new varieties in Pakistan. *Sci Total Environ* 371(1–3):304–313. <https://doi.org/10.1016/j.scitotenv.2006.06.017>
- World Health Organization (2014) Burden of disease from the joint effects of Household and Ambient Air Pollution for 2012. Public Health, Social and Environmental Determinants of Health Department, World Health Organization, 1211 Geneva 27, Switzerland
- Xiao J (2016) Chemical composition and source identification of rainwater constituents at an urban site in Xi'an. *Environ Earth Sci* 75(3):209. <https://doi.org/10.1007/s12665-015-4997-z>
- Xu G, Gao Y, Lin Q, Li W, Chen L (2013) Characteristics of water-soluble inorganic and organic ions in aerosols over the Southern Ocean and coastal East Antarctica during austral summer. *J Geophys Res* 118:13303–13318
- Zhang Y, Shao M, Zhang Y, Zeng L, He L, Zhu B, Wei Y, Zhu X (2007) Source profiles of particulate organic matters emitted from cereal straw burnings. *J Environ Sci* 19(2):167–175. [https://doi.org/10.1016/S1001-0742\(07\)60027-8](https://doi.org/10.1016/S1001-0742(07)60027-8)
- Zhang T, Engling G, Chan CY, Zhang YN, Zhang ZS, Lin M, Sang XF, Li YD, Li YS (2010) Contribution of fungal spores to particulate matter in a tropical rainforest. *Environ Res Lett* 5(2):024010. <https://doi.org/10.1088/1748-9326/5/2/024010>
- Zhang HH, Yang GP, Liu CY, LP S (2013) Chemical characteristics of aerosol composition over the Yellow Sea and the East China Sea in autumn. *J Atmos Sci* 70(6):1784–1794. <https://doi.org/10.1175/JAS-D-12-0232.1>
- Zhao Y, Gao Y (2008) Acidic species and chloride depletion in coarse aerosol particles in the US east coast. *Sci Total Environ* 407(1):541–547. <https://doi.org/10.1016/j.scitotenv.2008.09.002>